

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

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Index

- Introduction and Background
- Literature Survey
- Problem Statement
- Social Relevance and Uniqueness of the work
- Aim, Objective and Hypothesis
- Methodology
- References

Introduction and Background

- Public transport plays an important role to meet the daily needs of people across any Indian cities. In Mangalore, KSRTC buses are widely used by students, office commuters, inter-district passengers etc. but there is a need to address the challenges of overcrowding during certain days especially at peak hours as it affects comfort and planning.
- Most schedules do not give information about suitable travel timings or crowd levels. Also, existing studies focused only on real-time data, passenger counts, or automated systems which are not always accessible. As a result, there is a need for a simple approach that uses the publicly available data.
- This research uses KSRTC Mangalore bus timetables and district-level indicators to analyse the travel patterns and estimate the crowd risk pattern at different time slots. This can help passengers to choose the less congested timings without making real-time claims.

Literature Survey

Author & Journal Details	Titles of Articles & Objectives	Methodology	Outcomes of the Study	Relevance of Article to work
[1] . Basha, SK., et al. (2025)	AI-Driven Bus Boarding Demand Prediction for Real-Time Public Transport Optimization- To improve passenger boarding demand in bus transit by using SVM and DNNs, to forecast demand dynamically.	It involved training and validating models with timestamps, route numbers, and traffic conditions, and comparing performance.	AI-based models delivered high precision in predicting boarding demand.	Timetable and historical input features to estimate demand patterns, routing crowd estimation and travel time recommendation.
[2] . Shrivastava, A., et al. (2024)	Deep-Learning-Based Model for Prediction of Crowding in a Public Transit System- Develop a crowding prediction framework using ticketing data and clustering passenger segments.	Applied transit segment relations (TSR) to capture ridership patterns and predicts crowding levels for different periods of the day.	The model, validated with three months of ticketing data in Bhubaneswar, can guide passengers and transit agencies about expected crowding, helping them plan route choices.	Categorizing crowd levels based on time and ridership patterns, transforming into time slots and risk levels in timetable analysis without direct crowd counts.

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[3] . V S, D., et al. (2022)	Crowd Density Estimation and Location Prediction in Public Transport System– To design a system that combines crowd density estimation with location prediction.	Involved IoT, hardware sensors, and deep learning models to estimate crowd density and provide travel information.	Such systems can contribute to better use of travel time and help both passengers and travel agencies schedule appropriately.	It supports the use of crowd level estimation as a tool for travel planning using secondary data proxies.
[4] . Hsu, W. Y., et al. (2020)	Estimation of the Number of Passengers in a Bus Using Deep Learning- To estimate passengers by applying deep learning models that combine feature extraction and object detection.	It processed visual or sensor data to detect head regions and count passengers, evaluating performance across different scenarios.	The model performs better than existing methods in estimating occupancy.	Illustrating advanced crowd estimation techniques where accurate crowd estimation can be improved by using data patterns, from timetables and demographic indicators.

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[5] . Marques, S., et al. (2022)	Transit Ridership Modelling at the Bus Stop Level: Comparison of Approaches Focusing on Count and Spatially Dependent Data - Model boarding and alighting behaviour at the bus stop level.	Compared models with and without spatial dependence, finding that models recognizing spatial differences provide better fit and improve predictions.	Emphasize the importance of incorporating spatial context and demographic predictors into ridership estimates.	It validates the use of demographic and locational indicators (e.g., population, land use) for estimating crowd patterns from secondary data.
[6] . Pronello, C., et al. (2024)	A Low-Cost Automatic People-Counting System at Bus Stops Using Wi-Fi Probe Requests and Deep Learning-Estimate people waiting at bus stops, using Wi-Fi probe requests and CRNN.	Involved installing sniffing devices at stops, collecting probe request data, and training deep learning models to predict passenger counts with an average error of about 1.2 people.	Low-cost sensing combined with ML can yield reliable counts for planning.	It demonstrates modern crowd estimation techniques and provides a contrast between hardware-based crowd counting and timetable-based estimation.

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[7] . Talusan, JP., et al. (2022)	On Designing Day Ahead and Same Day Ridership Level Prediction Models Using Noisy APC Data- Design models for daily ridership levels with weather, transit, and calendar data.	XGBoost and LSTM was used on millions of observations to perform trip- and stop-level prediction, showing that data fusion and advanced models outperform baseline statistics.	Multi-source datasets improve accuracy.	Handling noisy secondary data and model training for feature engineering approach even without real ridership counts.
[8] . Martí, P., et al. (2024).	Bus Ridership Prediction and Scenario Analysis through ML and Multi-Agent Simulations - Combine ML forecasts & multi-agent simulations to model and analyse bus passenger flows and evaluate transport.	Comprehensive dataset and ML model training to forecast passenger flows, incorporating them into agent-based simulations to evaluate service performance and scenario outcomes.	ML predictions and agent-based simulations produces detailed analyses of ridership scenarios, offering insights on resource allocation, service efficiency, and optimisation of urban mobility.	Predictive modelling and scenario evaluation can inform crowd and demand estimation.

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[9] . Fontes, T., et al. (2020).	A Deep Learning Approach for Predicting Bus Passenger Demand Based on Weather Conditions-Predict bus ridership by incorporating weather conditions and time labels.	Transit bus ridership data and weather conditions over one year were used. A Multilayer Perceptron (deep learning regression model) was trained with time and weather features to predict passenger demand.	The inclusion of weather features (especially conditions two hours before travel) reduced prediction error by approximately 9%, and model sensitivity analysis.	Integrating contextual features (e.g., time of day and external indicators) to improve forecasting.
[10] . Farahmand, Z.H., et al. (2023)	Predicting Bus Ridership Based on the Weather Conditions Using Deep Learning Algorithms-Forecast bus ridership by incorporating weather conditions and others into demand prediction models.	Used smart-card fare data covering multiple years, and applied a Multilayer Perceptron under three scenarios: without weather data, with weather data at boarding time, and with weather data one hour ahead.	Delivered ridership forecasts, improving prediction accuracy when weather features were included. Error rates on weekdays and weekends improved.	Engineered external features (weather, time) can improve demand prediction.

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[11] . Yusuf, O., et al. (2025).	Data-Driven Predictive Modelling of Stop-Level Public Transit Patterns- Develop a predictive modelling framework that takes local spatiotemporal dynamics of passenger counts at stop level.	Horizon-agnostic machine learning framework using APC data, engineered features like boarding/alighting counts, operational and weather and land use, and many predictive algorithms with feature optimisation and interpretability analyses.	Tree-based ML models gave better passenger count forecasts. Enabled scenario analyses for planning applications and highlighted asymmetric relationships between boarding and alighting counts.	Structured feature engineering and machine learning can extract ridership patterns from high-resolution data.
[12] . Siswanto, J., et al. (2025)	Deep Learning Based LSTM Model for Predicting the Number of Passengers for Public Transport Bus Operators- Develop LSTM model to predict future bus passenger counts.	Time-series historical ridership data collected. Data pre-processing, feature selection and LSTM network training are performed with tuned parameters and evaluated with MSLE, MAPE, and SMAPE.	LSTM model performed well with optimized parameters, producing 12-month future predictions for each operator's ridership.	Time-series LSTM modelling can extract demand patterns and forecast passenger numbers.

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[13] . Drisilda, S., et al. (2025).	Machine Learning-Based Public Transit Ridership Forecasting System for BMTC-Develop ML system to forecast hourly ridership at BMTC bus stops using historical transit data.	Six months of historical ridership data from selected BMTC routes were analysed using a SARIMAX seasonal time-series model. The system captured daily and weekly seasonality and was deployed with an interactive forecasting interface.	SARIMAX achieved a mean absolute error of 18.92 passengers and a root mean squared error of 23.45, and supported real-time interactive visualization with route-level filtering.	Seasonal time-series models can predict ridership levels with reasonable accuracy using historic schedule data
[14] . NNS Amir, et al. (2025).	Ridership Prediction for Bus Rapid Transit Using Multi-Feature Machine Learning Approaches- Predict bus ridership levels by integrating operational, temporal, and contextual features.	Employed supervised ML trained on historical bus ridership data combined with time-based and route-level features. Multiple algorithms were evaluated and compared to identify the most accurate predictive framework for BRT systems.	Models with explanatory features significantly improved ridership prediction accuracy compared to baseline statistical methods. Clear peak-hour demand variations and route-specific crowding trends.	Supports feature-based crowd estimation, showing that even without real-time sensing, time, route, and service characteristics can meaningfully explain passenger demand patterns.

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Author & Journal Details	Titles of Articles & Objectives	Methodology	Outcomes of the Study	Relevance of Article to work
[15] . Movaghar, M., et al. (2025).	A Machine Learning Framework for Estimating Bus Ridership Loss During Network Disruptions-Estimate changes in bus ridership during disruptions or external events using ML.	Used historical ridership data and applied regression-based and tree-based machine learning models to quantify ridership deviations under different scenarios. Temporal patterns and route-level characteristics were included to model demand sensitivity.	ML models effectively captured ridership fluctuations caused by disruptions, with improved performance over traditional models and gave interpretable insights into how demand varies across routes and time periods.	Highlights the sensitivity of ridership to time-based and contextual factors, reinforcing that scheduled service characteristics and temporal segmentation (like time-slot framework) can be used to infer crowd risk levels.
[16] . Zeb, M. S., et al. (2024).	Forecasting Public Transport Ridership During Disruptive Periods Using LSTM Models-Analyse and forecast public transport ridership using deep learning techniques.	It used historical public transport ridership data and applied LSTM neural networks to capture temporal dependencies. The model performance was evaluated across pre-disruption, disruption, and recovery phases.	The LSTM model captured sharp declines and gradual recovery trends in ridership, outperforming traditional forecasting models during periods of high uncertainty.	Importance of time-based segmentation and temporal trends in understanding passenger behaviour.

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Author & Journal Details	Titles of Articles & Objectives	Methodology	Outcomes of the Study	Relevance of Article to work
[17] . Akgüngör, A. P., et al. (2025).	Comparison of LSTM and NARX Models for Short-Term Passenger Flow Prediction-Compare deep learning and nonlinear autoregressive models in predicting short-term passenger flows.	Applied LSTM and NARX to historical passenger flow data, incorporating temporal lags and trend features. Model accuracy was evaluated using standard error metrics to identify strengths and limitations of each approach.	LSTM models captured nonlinear and peak-hour passenger flow patterns, while NARX models were effective for stable demand periods.	Passenger demand varies significantly across time periods, validating decision to structure analysis around departure times and defined time slots for crowd risk estimation and travel recommendations.
[18] . Torrepadula, F. R. di., et al. (2024).	A Survey of Machine Learning Methods for Public Transport Demand Forecasting-Review and synthesise recent machine learning approaches.	Conducted a structured literature review of ML-based demand forecasting studies, comparing traditional statistical methods with machine learning and deep learning models across different transport modes and data types.	ML models outperform classical approaches when sufficient structured data is available, for capturing non-linear temporal patterns.	Theoretical grounding by showing that time-based and route-based features are widely used for demand estimation, supporting timetable-driven crowd estimation framework using secondary data.

Literature Survey

Author & Journal Details	Titles of Articles & Objectives	Methodology	Outcomes of the Study	Relevance of Article to work
[19] . Ke, L., et al. (2025).	UrbanBus: A Fine-Grained Dataset for Bus Ridership Analysis and Forecasting-Introduce a large-scale, fine-grained dataset to support bus ridership analysis, forecasting, and pattern discovery.	Compiled and processed millions of bus ridership records containing route, stop, and temporal attributes, and demonstrated how such data can be used for forecasting and analytical tasks.	Revealed strong daily and weekly ridership cycles, peak-hour congestion patterns, and route-specific demand variations, enabling more precise demand analysis.	Conceptually supports time-slot-based crowd pattern analysis, showing that structured temporal information alone can reveal meaningful crowd trends.
[20] . Yamamura, T., et al. (2023).	Bus Ridership Prediction Using Multiple LSTM Models with Trend Features-Improve bus ridership forecasting accuracy by combining LSTM models with trend and seasonality features.	Applied Long Short-Term Memory (LSTM) models to historical bus ridership time-series data, incorporating trend and seasonal decomposition to enhance prediction stability.	Improved prediction accuracy over single-model baselines and effectively captured peak and off-peak ridership variations.	Importance of temporal segmentation and trend-aware analysis, aligning with use of defined time slots to estimate relative crowd levels and recommend suitable travel timings.

Problem Statement

- People using KSRTC buses depend on schedules to plan their journey. Though these give clear information about the route and the timings, they do not give the safest time periods or the crowd levels. As a result, people have to struggle with peak-hour congestion and crowded travel times.
- Simultaneously, detailed passenger count data, real-time systems etc. are not publicly available, creating a gap between the required information for travel planning and the data accessible for analysis.
- There is a need for an approach where publicly available timetable data can be used to estimate the crowd levels and support time-based travel recommendations for KSRTC bus services in Mangalore.

Social Relevance and Uniqueness of the Work

- Public transport plays an important role in improving mobility, reducing traffic congestion, and ensuring access to daily services. However, overcrowding in buses often lead to discomfort and safety concerns.
- This research is socially relevant as it focuses on understanding and predicting bus passenger patterns using data-driven approaches, which can support better planning, improved service quality, and enhanced commuter experience.
- The uniqueness of this work lies in its timetable-driven and population-aware approach to understand crowd patterns. This research uses secondary data, making it low-cost, transparent, and suitable. By linking scheduled bus timings with district-level population and urbanisation indicators, the study provides a practical framework that can be easily adopted by transport authorities, researchers, and media platforms without privacy or operational concerns.

Aim, Objective and Hypothesis

Aim

To analyse KSRTC Mangalore bus timetables to estimate route-wise crowd levels and recommend optimal travel time slots for passengers, enabling them to avoid peak-hour congestion and make informed travel decisions.

Objective

- a) To structure KSRTC Mangalore bus timetable data in Microsoft Excel, taking origin, destination, service type, via place, and departure times.
- b) To classify departure times into predefined time slots and integrate district-level population and urbanisation indicators to estimate relative crowd levels.
- c) To develop a framework to assign overcrowding risk levels to different routes and time slots based on structured secondary data.
- d) To create a Tableau-based dashboard to visualise estimated crowd levels and recommend less congested travel time slots for passengers.
- e) To provide insights that are practical, transparent, and reproducible using publicly available data without depending on real-time measurements or sensitive operational information.

Aim, Objective and Hypothesis

Hypothesis

H_0 (Null Hypothesis): There is no significant variation in crowd levels across different KSRTC bus routes and time slots in Mangalore.

H_1 (Alternative Hypothesis): Crowd levels vary across different bus routes and time slots, and structured timetable data combined with population and urbanisation indicators can identify comparatively less crowded travel periods.

Sample Size

The research considers **all KSRTC Mangalore bus routes** listed in the official timetable, including both intra city, inter-state and inter-district services. Since the study uses **secondary data only**, every available route in the timetable is treated as part of the sample.

Procedure for data collection

- a) KSRTC Mangalore bus timetables will be collected from **official sources**, including bus stand notice boards, KSRTC website, and publicly available PDF schedules.
- b) Timetable data will be **structured in Microsoft Excel**, capturing the following fields: From, To, Service Type, Via Place, Departure Time.
- c) Departure times will be grouped into **predefined time slots** to reflect different periods of daily travel demand in 24-hour format.
- d) District-level **population data (in millions) and urbanisation levels** will be incorporated from official government statistics and mapped to routes based on service coverage.

Sources and Tools

Primary Data: Not applicable (no field observations or real-time measurements).

Secondary Data:

- a) KSRTC Mangalore official timetables (bus stand notices, PDF schedules, or website)
- b) Government census and urbanisation data for population indicators.

Microsoft Excel: For structuring data, categorisation, and initial calculations.

Python (Pandas, NumPy, optionally scikit-learn): For data cleaning, feature engineering, and classification of crowd risk levels.

Tableau: For creating an interactive dashboard that visualises route-wise crowd levels and recommends suitable travel time slots.

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